

Date: 02 July 2026	Team ID: SB-7429
Project Name: Credit Card Approval Prediction using Machine Learning	Maximum Marks: 5 Marks

PROJECT PROPOSAL (PROPOSED SOLUTION)

Project Proposal (Proposed Solution) Report

Project Overview:

Objective: Build an intelligent ML underwriting system that automates the screening process. By training predictive classifiers on historical consumer demographic and financial profiles, the system evaluates applicant parameters to instantly flag low-risk (Approved) and high-risk (Rejected) applications.

Scope: Relational merge of demographic and payment records, data cleaning, SMOTE oversampling, training/benchmarking, probability threshold calibration, construction of Layer 1 policy override filters, local compliance verification, and Flask web deployment.

Problem Statement:

Description: Traditional underwriting relies on rigid scoring or manual file review. This leads to operational inefficiencies, high decisioning latency, cognitive fatigue, human errors, and an inability to capture complex non-linear default relationships.

Impact: Resolving these issues reduces credit default risk (NPAs), eliminates manual review queues, minimizes customer acquisition costs (CAC), and allows underwriters to focus on complex accounts.

Proposed Solution Approach:

Implement a hybrid decision engine. The system first screens applicant attributes through 4 policy rules. If clear risk is detected, the application is short-circuited as a rejection. Otherwise, features are engineered and scaled, then processed by a pre-trained Random Forest model calibrated to a default probability threshold of 0.2312.

Resource Requirements Matrix

Resource Type	Component Description	Specification / Allocation
Hardware	Computing Resources	Local development workstation, Intel Core i5/i7 (4+ cores).
Hardware	System Memory	8 GB RAM minimum to run training partitions and Flask web server.
Hardware	System Storage	10 GB SSD space for raw CSV datasets, logs, and pickle binaries.
Software	Development Environment	VS Code / Jupyter Notebook / Git version control.
Software	Web Framework	Flask (2.0.2) web server, Jinja2 templating engine, Gunicorn.
Software	Machine Learning Stack	Scikit-Learn (Random Forest, Scaler), Pandas, NumPy, Imbalanced-Learn (SMOTE).
Data	Data Cache Source	Credit Card Approval Dataset (application_record.csv & credit_record.csv). ~36k rows in CSV format.